School of Innovation and Management (SoIM)
Post Graduate Diploma in Management (PGDM)
Business Analytics

Course Outline | **PGDM – BA (2025-27)** | **Trimester – IV**

Course Code: 404BA	Course Name: Data Mining & Unstructured Data Analysis	Course Credits: 3

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COURSE COVERAGE:

This course equips Business Analytics students with the concepts and hands-on skills to discover patterns and build predictive models from data using data mining, and to extract business insight from unstructured data such as text, web content, social media, networks, images, audio, and documents. It follows the CRISP-DM process — data understanding and exploration, preprocessing and feature engineering, dimensionality reduction, association rules, recommender systems, classification, ensemble methods, model evaluation, clustering, anomaly and fraud detection, customer analytics, and model interpretability — with attention to ethics, privacy, and bias. Working familiarity with Python and basic statistics is expected.

The second half of the course focuses on unstructured data analytics. Learners acquire data through web scraping and APIs, preprocess and represent text, and apply sentiment analysis, text classification, topic modeling, and word embeddings. They then explore social media and network analytics, image, audio, and video analytics, and Document AI (PDF and OCR). The course closes with Large Language Models (LLMs) and Generative AI for unstructured data, the integration of structured and unstructured data in predictive models, and the responsible use of data, culminating in a capstone project that solves a real business problem end to end.

COURSE OBJECTIVES:

By the end of this course, participants will be able to:

1. Frame business problems as data mining tasks using the CRISP-DM process and distinguish structured, semi-structured, and unstructured data.
2. Explore, clean, transform, and engineer features from raw data, and reduce dimensionality using techniques such as PCA.
3. Discover patterns in transactional data using association rule mining (Apriori, FP-Growth) and build basic recommender systems.
4. Build classification models using decision trees, k-Nearest Neighbors, Naïve Bayes, Logistic Regression, and ensemble methods.

5. Evaluate and validate models using confusion matrices, precision, recall, F1-score, ROC-AUC, and cross-validation, while handling class imbalance.
6. Segment customers and data using k-Means, hierarchical, and density-based clustering, and detect anomalies and fraud.
7. Apply RFM analysis and churn prediction to solve customer analytics problems, and interpret models using feature importance, SHAP, and LIME.
8. Acquire unstructured data through web scraping and APIs, and preprocess text using tokenization, stop-word removal, stemming, lemmatization, and regular expressions.
9. Represent text using Bag of Words, TF-IDF, and n-grams, and perform sentiment analysis, text classification, and document clustering.
10. Discover themes in large text collections using topic modeling (LDA, NMF) and measure semantic similarity using word and sentence embeddings.
11. Analyze social media data and networks using centrality and community-detection measures to identify trends and influencers.
12. Describe and apply basic techniques for image, audio, video, and document (PDF and OCR) analytics.
13. Use LLMs and Generative AI for summarization, information extraction, and semantic search, and integrate structured and unstructured data into business insights while applying privacy, ethics, and data-governance principles.

COURSE LEARNING OUTCOMES:

After completing the course, students will be able to demonstrate proficiency in:

CO1 – Data Mining Foundations & Data Preparation:

Frame business problems using CRISP-DM; explore, clean, transform, and engineer features from structured and semi-structured data; and apply dimensionality reduction to prepare data for mining.

CO2 – Pattern Discovery: Associations, Clusters & Anomalies:

Discover patterns using association rules, recommender techniques, clustering (k-Means, hierarchical, DBSCAN), and anomaly detection, and apply them to market-basket, customer-segmentation, and fraud-detection problems.

CO3 – Predictive Modeling, Evaluation & Interpretation:

Build classification models using decision trees, k-NN, Naïve Bayes, Logistic Regression, and ensembles; evaluate them using appropriate metrics and validation; and interpret and communicate model results responsibly.

CO4 – Text & Web Analytics:

Acquire data through web scraping and APIs; preprocess and represent text; and apply sentiment analysis, text classification, topic modeling, and embeddings to extract insight from reviews, feedback, and social media content.

CO5 – Network, Multimedia & Generative AI Analytics:

Analyze networks, images, audio, video, and documents; use LLMs and Generative AI for unstructured data; integrate structured and unstructured data into decision-ready insights; and apply privacy and ethical principles in a capstone project.

ASSESSMENT TOOLS FOR LEARNING OUTCOME:

Learning Outcome	Assessment Criteria
Frame business problems using CRISP-DM and explore data using EDA	5%
Clean, transform, and engineer features from raw data; apply dimensionality reduction	10%
Discover patterns using association rules and recommender techniques	5%
Build and tune classification models (trees, k-NN, Naïve Bayes, Logistic Regression, ensembles)	15%
Evaluate models using confusion matrix, ROC-AUC, and cross-validation; interpret with feature importance and SHAP	10%
Segment data using clustering and detect anomalies and fraud	10%
Acquire unstructured data via web scraping and APIs; preprocess and represent text	10%
Perform sentiment analysis, text classification, and topic modeling	10%
Apply network, image, audio, and document analytics techniques	5%
Use LLMs and embeddings for summarization, information extraction, and semantic search	5%
Capstone project: an end-to-end analytics solution integrating structured and unstructured data	15%

PEDAGOGY:

- **Professor-led lectures and live coding demonstrations** to introduce core concepts in data mining, predictive modeling, text analytics, and unstructured data analysis.
- **Hands-on Lab sessions** in Python (Jupyter Notebook / Google Colab) using real-world business datasets — retail transactions, telecom churn, credit-card fraud, product reviews, and social media posts — to develop practical skills.

- **Skills Application Lab** session where students independently apply previously taught concepts to solve structured problem sets, consolidating data mining and text analytics skills before moving to advanced unstructured data topics.
- **Guided practice and in-class activities** with knowledge checks to ensure students build, evaluate, and interpret models independently.
- **Project work** involves a capstone project that brings together data mining and unstructured data analytics to solve a real business problem, from problem framing to insight communication.
- **Peer review activities** for feedback on notebooks and capstone presentations, encouraging reflection on modeling choices, evaluation rigor, and clarity of business recommendations.
- **Assignment** to assess mastery of core techniques such as preprocessing, classification, clustering, model evaluation, and text analytics.
- **Case studies based on business scenarios** provide authentic contexts, helping students see how data mining and unstructured data analytics support marketing, risk, operations, and customer-experience decisions.
- **Continuous feedback and support from the professor** to drive skill improvement across all learning outcomes, ensuring students are ready for practical application in analytics roles.

SESSION PLAN:

20 sessions of 90 minutes each, totalling 30 Hours for this 3-credit course.

Session Plan			Pedagogy
Sl. No.	**Topic**	**Pre-reading / Reference**	
1	**Introduction:** • What is Data Mining • Motivating Challenges • Origins of Data mining • Data Mining Tasks **Data:** • Type • Data Quality • Pre-processing	Chapter 1 & 2 Introduction to Data Mining (Tan, P.)	• Lecture and discussion • Case discussion: data mining in retail and banking

Session Plan			Pedagogy
Sl. No.	**Topic**	**Pre-reading / Reference**	
2	**Data:** • Measures of Similarity and Dissimilarity **Classification:** • Basic concepts and framework • Decision Tree classifier Model Overfitting **Data Preprocessing & Cleaning** • Missing values, duplicates, outlier treatment, encoding, scaling, train-test split	Han et al. – Data Mining: Concepts and Techniques: Getting to Know Your Data; Data Preprocessing	• Lecture and live demo • Lab: EDA and cleaning of a retail dataset
3	**Feature Engineering & Selection** • Derived features, binning, feature importance, filter and wrapper methods **Dimensionality Reduction** • Curse of dimensionality, PCA, explained variance, visualization	Han et al. – Data Preprocessing (data reduction) Leskovec et al. – Mining of Massive Datasets: Dimensionality Reduction scikit-learn User Guide: Decomposition (PCA)	• Lecture and live coding demo • Lab: feature engineering and PCA
4	**Market Basket Analysis & Association Rules** • Itemsets, support, confidence, lift, Apriori, FP-Growth **Recommender Systems** • Content-based filtering, collaborative filtering, similarity, cold-start problem	Leskovec et al. – Mining of Massive Datasets: Frequent Itemsets; Recommendation Systems mlxtend documentation: apriori and association_rules	• Lab: Apriori on retail transactions • Exercise: build a simple recommender
5	**Decision Trees & k-Nearest Neighbors** • Entropy, Gini index, tree pruning, distance measures, choice of k **Naïve Bayes & Logistic Regression** • Conditional probability, odds, probability threshold, classification for business decisions	Provost & Fawcett – Ch. 3 (Predictive Modeling) and Ch. 6 (Similarity, Neighbors, and Clusters) Ch. 9 (Evidence and Probabilities)	• Lecture and live coding demo • Lab: classify loan applicants
6	**Ensemble Methods** • Bagging, random forest, boosting, gradient boosting, feature importance **Model Evaluation & Validation** • Confusion matrix, precision, recall, F1-score, ROC-AUC, cross-validation, class imbalance	Géron – Hands-On Machine Learning: Ensemble Learning and Random Forests Provost & Fawcett – Ch. 7 and Ch. 8	• Lab: random forest and gradient boosting • Assignment 1: classification model with evaluation
7	**Clustering: k-Means & Hierarchical** • Centroids, choosing k, elbow method, dendrogram, linkage **Density-Based Clustering & Cluster Validation** • DBSCAN, silhouette score, cluster profiling, customer segmentation	Leskovec et al. – Mining of Massive Datasets: Clustering scikit-learn User Guide: Clustering	• Lab: customer segmentation with k-Means and DBSCAN • Guided practice

Session Plan			Pedagogy
Sl. No.	**Topic**	**Pre-reading / Reference**	
8	**Anomaly & Fraud Detection** • Outliers, statistical methods, isolation forest, fraud detection use cases **Customer Analytics** • RFM analysis, segmentation, churn prediction, customer lifetime value	Han et al. – Outlier Detection scikit-learn User Guide: Novelty and outlier detection	• Lab: isolation forest on card transactions • Case study: RFM segmentation and churn
9	**Model Interpretability** • Feature importance, partial dependence, SHAP, LIME, explaining models to stakeholders **Ethics, Privacy & Bias in Data Mining** • Fairness, algorithmic bias, privacy, data protection, responsible use	Molnar – Interpretable Machine Learning (online book)	• Lab: SHAP explanations • Discussion: bias in credit scoring • Quiz 1
10	**End-to-End Data Mining Case Study** • Business problem framing, data preparation, modeling, evaluation, recommendations, deployment considerations	Provost & Fawcett – Ch. 2 (Business Problems and Data Science Solutions)	• Case study workshop • Team presentations • Mid-term review
MID-TERM EXAMINATION			
11	**Introduction to Unstructured Data Analytics** • Unstructured data sources, challenges, business value, analytics pipeline **Web Scraping & APIs** • HTML structure, requests, BeautifulSoup, JSON APIs, scraping ethics	Mitchell – Web Scraping with Python BeautifulSoup and requests documentation	• Lecture and discussion • Lab: scrape product reviews and call a JSON API
12	**Text Preprocessing & Regular Expressions** • Cleaning, tokenization, stop words, stemming, lemmatization, regex **Text Representation** • Bag of Words, TF-IDF, n-grams, document-term matrix, cosine similarity	Bird et al. – NLTK Book, Ch. 3 (Processing Raw Text) Manning et al. – Introduction to Information Retrieval: Scoring, term weighting and the vector space model	• Lab: text cleaning and regex • Lab: TF-IDF matrices and cosine similarity
13	**Sentiment & Opinion Mining** • Lexicon-based sentiment (VADER, TextBlob), ML-based sentiment, review analytics, aspect-level sentiment **Text Classification & Document Clustering** • Spam detection, category tagging, document clustering, evaluation	Hutto & Gilbert (2014) – VADER Bird et al. – NLTK Book, Ch. 6 (Learning to Classify Text)	• Lab: sentiment scoring of product reviews • Assignment 2: text classifier

Session Plan			Pedagogy
Sl. No.	**Topic**	**Pre-reading / Reference**	
14	**Topic Modeling** • LDA, NMF, topic coherence, topic interpretation, customer-feedback themes **Word Embeddings & Semantic Search** • Word2Vec, GloVe, sentence embeddings, semantic similarity, semantic search	Blei, Ng & Jordan (2003) – Latent Dirichlet Allocation gensim documentation	• Lab: LDA on customer feedback • Demo: sentence embeddings and semantic search • Quiz 2
15	**Social Media Analytics** • Social media data, hashtags, mentions, trend analysis, brand monitoring **Network Analysis** • Nodes and edges, centrality, community detection, influencers, NetworkX	Easley & Kleinberg – Networks, Crowds, and Markets NetworkX documentation	• Lab: hashtag and trend analysis • Lab: influencer network with NetworkX
16	**SKILLS APPLICATION LAB** *Independent, structured problem sets integrating data preprocessing, classification, clustering, association rules, and text analytics (Sessions 1–15).*		
17	**Image Data Analytics** • Pixels and channels, image preprocessing, feature extraction, CNN basics, transfer learning **Audio & Video Analytics** • Speech-to-text, audio features, call-centre analytics, video frame analysis	Géron – Hands-On Machine Learning: Deep Computer Vision Using Convolutional Neural Networks Hugging Face Audio Course	• Lecture and live demo • Lab: image classification with a pre-trained model • Demo: speech-to-text
18	**Document AI: PDF & OCR** • Text extraction from PDFs, OCR, invoices, tables, information extraction **LLMs & Generative AI for Unstructured Data** • Prompting, summarization, information extraction, embeddings, retrieval-augmented generation (RAG) basics	Tesseract / pytesseract documentation Tunstall et al. – Natural Language Processing with Transformers: Hello Transformers	• Lab: extract text and tables from PDFs • Lab: LLM-based information extraction
19	**Integrating Structured & Unstructured Data** • Feature fusion, text features in predictive models, dashboards, insight storytelling **Responsible Use of Unstructured Data** • Privacy, consent, Digital Personal Data Protection Act 2023, bias, data governance	Provost & Fawcett – Ch. 13 (Data Science and Business Strategy) Digital Personal Data Protection Act, 2023 (Government of India)	• Lab: add text features to a churn model • Assignment 3: insight storytelling • Quiz 3
20	**Capstone Project Presentations & Course Review** • Problem framing, structured and unstructured data pipeline, insights, recommendations, peer feedback	Course notebooks and capstone guidelines	• Capstone presentations • Peer review and feedback • Course review
END-TERM EXAMINATION			

RECOMMENDED TEXTBOOK:

Han, J., Pei, J., & Tong, H. (2022). *Data mining: Concepts and techniques* (4th ed.). Morgan Kaufmann.

REFERENCE TEXTBOOKS/RESOURCES:

1. Provost, F., & Fawcett, T. (2013). *Data science for business: What you need to know about data mining and data-analytic thinking*. O'Reilly Media.
2. Tan, P.-N., Steinbach, M., Karpatne, A., & Kumar, V. (2019). *Introduction to data mining* (2nd ed.). Pearson.
3. Géron, A. (2022). *Hands-on machine learning with Scikit-Learn, Keras, and TensorFlow* (3rd ed.). O'Reilly Media.
4. Leskovec, J., Rajaraman, A., & Ullman, J. D. (2020). *Mining of massive datasets* (3rd ed.). Cambridge University Press. http://www.mmds.org/
5. Bird, S., Klein, E., & Loper, E. (2009). *Natural language processing with Python*. O'Reilly Media. https://www.nltk.org/book/
6. Manning, C. D., Raghavan, P., & Schütze, H. (2008). *Introduction to information retrieval*. Cambridge University Press. https://nlp.stanford.edu/IR-book/
7. Mitchell, R. (2018). *Web scraping with Python* (2nd ed.). O'Reilly Media.
8. Easley, D., & Kleinberg, J. (2010). *Networks, crowds, and markets: Reasoning about a highly connected world*. Cambridge University Press.
9. Molnar, C. (2022). *Interpretable machine learning* (2nd ed.). https://christophm.github.io/interpretable-ml-book/
10. Tunstall, L., von Werra, L., & Wolf, T. (2022). *Natural language processing with transformers* (Rev. ed.). O'Reilly Media.
11. scikit-learn documentation: https://scikit-learn.org/stable/user_guide.html

COURSE EVALUATION:

Evaluation is a continuous process at SoIM. Students will be assigned coding assignments, quizzes, and projects aligned with the course learning outcomes. The evaluation scheme is as follows:

Areas	Weightage
(I) Continuous Evaluation: (Quizzes: 10%; Assignments and Group/Individual Projects: 30%)	40%
(II) Mid Term Examination	20%
(III) Final Examination	40%
Total	**100 %**

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